

# CHRONIC KIDNEY DISEASE

## Clinical Management, Treatment Protocols & Pharmacotherapy Guide

Nephrology Clinical Reference Series | 3rd Edition, 2025

~10%

Global Prevalence

~800M

People Affected

50%+

Cases Undiagnosed

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SECTION 1 — OVERVIEW, EPIDEMIOLOGY & STAGING

Chronic Kidney Disease (CKD) is defined as abnormalities of kidney structure or function, present for more than 3 months, with implications for health. CKD encompasses a spectrum of conditions characterised by progressive loss of kidney function, ultimately leading to end-stage kidney disease (ESKD). The disease is a major public health burden, affecting approximately 9.1% of the global population and representing one of the leading causes of morbidity and premature mortality worldwide.

The two most common causes of CKD are diabetic nephropathy (accounting for 30–40% of all cases) and hypertensive nephrosclerosis (25–30%). Other aetiologies include glomerulonephritis, polycystic kidney disease (PKD), lupus nephritis, IgA nephropathy, obstructive uropathy, and recurrent urinary tract infections with parenchymal damage.

1.1 KDIGO Staging System

The Kidney Disease: Improving Global Outcomes (KDIGO) classification stratifies CKD by glomerular filtration rate (GFR) category and albuminuria category, producing a heat-map risk matrix that guides clinical decision-making and monitoring frequency.

| GFR Category | eGFR (mL/min/1.73 m <sup>2</sup> ) | Description                                 | Clinical Action                  |
|--------------|------------------------------------|---------------------------------------------|----------------------------------|
| G1           | ≥ 90                               | Normal/High — Kidney damage with normal GFR | Treat cause; CVD risk reduction  |
| G2           | 60–89                              | Mildly decreased                            | Treat cause; slow progression    |
| G3a          | 45–59                              | Mildly to moderately decreased              | Nephrology referral consider     |
| G3b          | 30–44                              | Moderately to severely decreased            | Nephrology co-management         |
| G4           | 15–29                              | Severely decreased                          | RRT planning, dietitian referral |
| G5           | < 15                               | Kidney failure (ESKD)                       | Initiate dialysis or transplant  |

Table 1. KDIGO GFR Categories and Clinical Actions

1.2 Albuminuria Classification (ACR)

| Category | ACR (mg/g or mg/mmol)      | Description                                         |
|----------|----------------------------|-----------------------------------------------------|
| A1       | < 30 mg/g (< 3 mg/mmol)    | Normal to mildly increased                          |
| A2       | 30–300 mg/g (3–30 mg/mmol) | Moderately increased (microalbuminuria)             |
| A3       | > 300 mg/g (> 30 mg/mmol)  | Severely increased (macroalbuminuria / proteinuria) |

Table 2. KDIGO Albuminuria Categories

## SECTION 2 — DIAGNOSTIC WORKUP & CLINICAL ASSESSMENT

Comprehensive evaluation of CKD requires an integrated approach incorporating laboratory markers, imaging, and targeted biopsy when indicated. Early identification of underlying aetiology is critical as it determines specific treatment pathways and prognostic trajectory.

### 2.1 Essential Laboratory Panel

- **Kidney Function:** Serum creatinine, BUN, eGFR (CKD-EPI 2021 equation recommended), cystatin C for borderline cases
- **Albuminuria:** Spot urine albumin-to-creatinine ratio (ACR); 24-hour urine protein only if ACR unreliable
- **Electrolytes:** Sodium, potassium, bicarbonate, chloride; calcium, phosphate, magnesium
- **Haematology:** CBC with differential; reticulocyte count if anaemia suspected; iron stores (serum iron, TIBC, ferritin)
- **Endocrine/Metabolic:** PTH (intact), 25-OH Vitamin D, HbA1c (if diabetic), fasting lipid panel
- **Urine Microscopy:** Red cell casts (glomerulonephritis), waxy casts (advanced CKD), granular casts
- **Serology (if indicated):** ANA, dsDNA, ANCA, anti-GBM, complement C3/C4, serum protein electrophoresis

### 2.2 Imaging Protocols

- **Renal Ultrasound (first-line):** Assess kidney size (normal 9–12 cm), cortical thickness, echogenicity, hydronephrosis, cysts, masses
- **CT Urography:** Obstructive uropathy, renal calculi, structural anomalies
- **MRI:** Preferred when contrast risk high; fibrosis assessment with newer diffusion-weighted sequences
- **DMSA Scan:** Cortical scarring assessment, particularly in reflux nephropathy

### 2.3 Kidney Biopsy Indications

Percutaneous kidney biopsy is indicated when the underlying cause is uncertain and may alter management, including: unexplained rapid GFR decline ( $>5$  mL/min/year), nephrotic-range proteinuria without apparent cause, glomerular haematuria, suspected vasculitis or systemic disease affecting kidneys, or atypical presentation of diabetic nephropathy. Contraindications include solitary kidney (relative), uncontrolled hypertension, bleeding diathesis, and uncooperative patient.

## SECTION 3 — PHARMACOTHERAPY & RENOPROTECTIVE AGENTS

The pharmacological management of CKD has undergone a paradigm shift over the past decade. While ACE inhibitors and ARBs have long been the cornerstone of renoprotective therapy, the emergence of SGLT2 inhibitors and non-steroidal mineralocorticoid receptor antagonists (nsMRAs) represents the most significant advance in CKD pharmacotherapy in 25 years. A comprehensive cardiorenal risk reduction approach is now recommended across all CKD stages.

### 3.1 First-Line Renoprotective Therapy

| Drug Class            | Agent / Dose                                                                                                 | Indication                                                          | Key Monitoring                                                                      | Contraindications                                                             |
|-----------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| ACE Inhibitor         | Ramipril 5–10 mg OD<br>Lisinopril 10–40 mg OD<br>Enalapril 5–20 mg BD                                        | Diabetic nephropathy;<br>proteinuric CKD G1-G4;<br>hypertension     | K+, creatinine at 1–2 wks;<br>SCr rise <30% acceptable                              | Bilateral RAS;<br>K+ > 5.5; pregnancy;<br>concurrent ARB                      |
| ARB                   | Losartan 50–100 mg OD<br>Irbesartan 150–300 mg OD<br>Olmesartan 20–40 mg OD                                  | ACE-inhibitor intolerant;<br>diabetic nephropathy; heart<br>failure | As per ACEi; avoid dual<br>RAAS blockade                                            | Same as ACEi;<br>angio-oedema history<br>with ACEi (relative)                 |
| SGLT2 Inhibitor       | Dapagliflozin 10 mg OD<br>Empagliflozin 10–25 mg<br>OD<br>Canagliflozin 100 mg OD                            | CKD G2-G4 + proteinuria<br>(eGFR ≥ 25); T2DM; heart<br>failure      | eGFR (hold if <20); UTI<br>symptoms; DKA signs;<br>genital mycosis                  | eGFR < 20; T1DM;<br>recurrent UTI; on loop<br>diuretics (caution)             |
| nsMRA<br>(Finerenone) | Finerenone 10–20 mg OD<br>(Start 10 mg if eGFR<br>25–60)                                                     | T2DM + CKD with<br>albuminuria despite RAAS<br>blockade             | K+ (monthly x3, then<br>quarterly); eGFR; avoid<br>with strong CYP3A4<br>inhibitors | K+ > 5.0 at initiation;<br>severe hepatic<br>impairment; Addison's<br>disease |
| GLP-1 RA              | Semaglutide 0.5–2 mg SC<br>weekly<br>Liraglutide 1.2–1.8 mg SC<br>OD<br>Dulaglutide 0.75–1.5 mg<br>SC weekly | T2DM with CKD;<br>cardiorenal risk reduction;<br>obesity            | Renal function; GI<br>tolerability; pancreatitis<br>symptoms                        | MEN2;<br>personal/family<br>history medullary<br>thyroid carcinoma            |

Table 3. Evidence-Based Pharmacotherapy for CKD Renoprotection

■ **CLINICAL ALERT:** Combination of ACEi + ARB is contraindicated due to increased risk of hyperkalaemia, acute kidney injury, and hypotension without additional benefit (ONTARGET trial). SGLT2 inhibitors provide renoprotection independent of glycaemic control; benefit extends to non-diabetic CKD.

### 3.2 Blood Pressure Targets & Antihypertensive Protocol

- **Target BP:** < 120/80 mmHg (SPRINT, ACCORD trials); individualise in elderly or frail patients
- **First-line:** ACEi or ARB (mandatory in proteinuric CKD) — titrate to maximum tolerated dose
- **Second-line:** Calcium channel blocker (amlodipine 5–10 mg OD) or thiazide-like diuretic (chlorthalidone preferred over HCTZ in CKD G1-G3b)
- **Loop diuretics:** Furosemide 20–80 mg OD/BD for volume overload in CKD G4-G5; torsemide preferred (superior oral bioavailability)
- **Resistant hypertension:** Add spironolactone 25 mg OD (monitor K+), then consider alpha-blockers or beta-blockers

## SECTION 4 — COMPLICATION MANAGEMENT PROTOCOLS

### 4.1 CKD-Mineral and Bone Disorder (CKD-MBD)

CKD-MBD is a systemic disorder manifest by abnormalities in calcium, phosphate, PTH, Vitamin D metabolism, and bone/vascular pathology. It begins in CKD G3 and becomes near-universal in G4-G5. FGF-23 elevation is the earliest biochemical marker, preceding hyperphosphataemia and secondary hyperparathyroidism.

- **Phosphate control (target: 0.8–1.5 mmol/L):** Dietary phosphate restriction 800–1000 mg/day; phosphate binders — calcium carbonate 500–1500 mg with meals (avoid in vascular calcification), sevelamer hydrochloride 800–1600 mg TDS (preferred in G4-G5), lanthanum carbonate 750–3000 mg/day
- **Vitamin D supplementation:** Correct 25-OH Vitamin D deficiency (target > 30 ng/mL) with cholecalciferol before activating analogues; activated Vitamin D (calcitriol 0.25–0.5 mcg OD or alfacalcidol 0.25–1 mcg OD) for secondary hyperparathyroidism
- **PTH targets by GFR category:** G3 — maintain within normal range; G4 — up to 2x upper limit; G5D (dialysis) — 2–9x upper limit of normal
- **Cinacalcet:** 30–180 mg OD for tertiary/refractory secondary hyperparathyroidism (dialysis patients); calcimimetic action; monitor hypocalcaemia

### 4.2 Anaemia of CKD

Anaemia in CKD is primarily due to reduced erythropoietin (EPO) production. Target haemoglobin is 10–12 g/dL; avoid >13 g/dL (increased CV events, TREAT trial). Iron deficiency must be corrected before ESA therapy.

- **Iron therapy:** Oral ferrous sulphate 200 mg BD-TDS (if tolerated); IV iron sucrose 200 mg over 30 min or ferric carboxymaltose 500–1000 mg (preferred in advanced CKD or ESA hyporesponsiveness). Target: ferritin 200–500 ng/mL, TSAT 20–30%
- **Erythropoiesis-Stimulating Agents (ESAs):** Epoetin alfa 50–300 units/kg SC 3x/week; darbepoetin alfa 0.45–0.75 mcg/kg SC every 1–4 weeks (longer half-life, less frequent dosing)
- **HIF-PHI (Hypoxia-Inducible Factor Prolyl Hydroxylase Inhibitors):** Roxadustat 70–200 mg 3x/week; daprodustat 1–24 mg OD — oral agents approved for anaemia in non-dialysis CKD; avoid in active malignancy

### 4.3 Hyperkalaemia Management

Hyperkalaemia ( $K^+ > 5.5$  mmol/L) is common in CKD G3-G5, exacerbated by RAAS blockade, diabetes, and acidosis. New potassium binders have revolutionised management, enabling continuation of life-saving RAAS blockade.

| K <sup>+</sup> Level                    | Management Protocol                                                                                                                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| K <sup>+</sup> 5.0–5.5                  | Dietary review; ensure no contributing drugs; repeat in 1–2 weeks                                                                                         |
| K <sup>+</sup> 5.5–6.0                  | Dietary restriction to <2g K <sup>+</sup> /day; consider patiomer or SZC; review RAAS dose                                                                |
| K <sup>+</sup> 6.0–6.5 (no ECG changes) | Patiomer 8.4 g OD with food OR sodium zirconium cyclosilicate (SZC) 10 g TDS x48h then 5 g OD                                                             |
| K <sup>+</sup> > 6.5 or ECG changes     | EMERGENCY: IV calcium gluconate 10% 10 mL; insulin-dextrose (10 units soluble + 50 mL 50% glucose); salbutamol 10–20 mg nebulised; dialysis if refractory |

Table 4. Hyperkalaemia Stepwise Management in CKD

### 4.4 Metabolic Acidosis

- **Target:** Serum bicarbonate  $\geq 22$  mmol/L; low bicarb accelerates CKD progression (BICAR-K study)
- **Treatment:** Sodium bicarbonate 500 mg–1 g TDS, titrate to serum bicarb; sodium citrate as alternative; caution with sodium load in hypertension/oedema

- **Dietary:** Mediterranean or plant-predominant diet (lower acid load, higher alkali content)

## SECTION 5 — RENAL REPLACEMENT THERAPY

Renal replacement therapy (RRT) should be discussed proactively when eGFR falls below 20 mL/min/1.73 m<sup>2</sup> to allow adequate time for patient education, modality selection, and access creation. The choice between haemodialysis (HD), peritoneal dialysis (PD), and kidney transplantation is patient-centred, guided by clinical factors, lifestyle, comorbidities, and patient preference.

### 5.1 Indications for Urgent Dialysis Initiation (AEIOU Mnemonic)

- **A — Acidosis:** pH < 7.1 or bicarbonate < 8 mmol/L refractory to medical management
- **E — Electrolytes:** Refractory hyperkalaemia (K<sup>+</sup> > 6.5 mmol/L with ECG changes) or severe hyponatraemia
- **I — Intoxication:** Dialysable toxins — salicylate, lithium, methanol, ethylene glycol, metformin-associated lactic acidosis
- **O — Overload:** Pulmonary oedema refractory to diuretic therapy; severe fluid overload
- **U — Uraemia:** Uraemic encephalopathy, pericarditis, pleuritis, severe nausea/vomiting, bleeding diathesis

### 5.2 Haemodialysis Parameters

- **Adequacy:** Single-pool Kt/V ≥ 1.2 per session (minimum); URR > 65%
- **Standard dose:** 3 sessions/week, 4 hours each; home HD (6x/week nocturnal HD) superior in BP control and phosphate management
- **Vascular access:** Arteriovenous fistula (AVF) is preferred (Brescia-Cimino); AVG if vessel unsuitable; tunnelled central line as temporary access only
- **Anticoagulation:** Unfractionated heparin (bolus + infusion) standard; low molecular weight heparin for circuits; citrate anticoagulation for HIT

### 5.3 Peritoneal Dialysis

- **Modalities:** CAPD (4 exchanges x 2L/day) or APD/CCPD using cycler overnight — preferred for residual renal function preservation
- **Adequacy:** Total weekly Kt/V ≥ 1.7; includes residual kidney contribution
- **Complications:** Peritonitis (Staph. epidermidis most common; treat with intraperitoneal antibiotics), catheter exit-site infections, PD ultrafiltration failure
- **Dialysate options:** Standard glucose 1.36%/2.27%/3.86%; icodextrin (7.5%) for long dwells or ultrafiltration failure; amino acid solutions for nutritional support

## SECTION 6 — DRUG DOSING ADJUSTMENTS IN CKD

Impaired renal clearance necessitates dose adjustment of renally eliminated drugs to avoid accumulation and toxicity. eGFR-based dose modification is standard, though drug-specific pharmacokinetic data should always be consulted (BNF, Renbase, Dialysis of Drugs).

| Drug Class / Agent      | eGFR 30–60                   | eGFR 15–30                   | eGFR < 15 / HD                    |
|-------------------------|------------------------------|------------------------------|-----------------------------------|
| Metformin               | Reduce dose; check 3-monthly | AVOID — lactic acidosis risk | CONTRAINDICATED                   |
| Metoprolol              | No adjustment                | Reduce by 25–50%             | Reduce by 50%; supplement post-HD |
| Gabapentin / Pregabalin | Reduce dose by 50%           | Reduce by 75%                | 200 mg post-HD only               |
| Amoxicillin             | Standard dose                | 500 mg BD (reduce frequency) | 250–500 mg OD; supplement post-HD |

| Drug Class / Agent | eGFR 30–60                                  | eGFR 15–30                         | eGFR < 15 / HD                   |
|--------------------|---------------------------------------------|------------------------------------|----------------------------------|
| Digoxin            | 0.0625–0.125 mg OD                          | 0.0625 mg OD; monitor levels       | AVOID or 0.0625 mg every 48h     |
| Enoxaparin (LMWH)  | Standard; consider anti-Xa monitoring       | Unfractionated heparin preferred   | UFH only; LMWH accumulates       |
| Vancomycin         | Extended interval (48–72h); monitor troughs | 72–96h interval; AUC-guided dosing | 3rd dose post-HD; AUC/MIC dosing |
| NSAIDs             | Avoid if possible; short course only        | AVOID                              | ABSOLUTELY CONTRAINDICATED       |
| Allopurinol        | 100–200 mg OD (start 50 mg)                 | 50–100 mg OD                       | 50 mg OD; supplement post-HD     |

Table 5. Common Drug Dose Adjustments by CKD Stage (eGFR-based)

■ NSAIDs are nephrotoxic and should be avoided in all stages of CKD unless absolutely necessary for <72 hours with adequate hydration and close monitoring. Contrast media: use iso-osmolar iodinated contrast; pre-hydrate with 0.9% saline; hold metformin and nephrotoxins 48h peri-procedure.

## SECTION 7 — MONITORING & FOLLOW-UP PROTOCOLS

Structured monitoring is essential for early detection of CKD progression, complication management, and treatment optimisation. The frequency of follow-up is determined by GFR category, albuminuria category, and clinical complexity.

| Parameter         | CKD G1-G2      | CKD G3            | CKD G4           | CKD G5/Dialysis                   |
|-------------------|----------------|-------------------|------------------|-----------------------------------|
| eGFR/Creatinine   | Annually       | Every 6 months    | Every 3 months   | Monthly                           |
| Urine ACR         | Annually       | Every 6 months    | Every 3 months   | Monthly                           |
| Blood Pressure    | Every visit    | Every visit       | Every visit      | Pre/post every session            |
| Potassium         | Annually       | Every 6 months    | Every 3 months   | Monthly (pre-HD x3/wk)            |
| Bicarbonate       | Annually       | Every 6 months    | Every 3 months   | Monthly                           |
| Calcium/Phosphate | Annually       | Every 6–12 months | Every 3 months   | Monthly                           |
| PTH               | If abnormal    | Every 6–12 months | Every 3–6 months | Every 3 months                    |
| Haemoglobin       | Annually       | Every 6 months    | Every 3 months   | Monthly                           |
| HbA1c (T2DM)      | Every 3 months | Every 3 months    | Every 3–6 months | Every 3 months (unreliable in G5) |
| Lipid Profile     | Annually       | Annually          | Annually         | Annually                          |

Table 6. CKD Monitoring Frequency by Disease Stage

### 7.1 Nephrology Referral Criteria

- eGFR < 30 mL/min/1.73m<sup>2</sup> (G4-G5) — mandatory; ideally when eGFR < 45 for RRT planning
- Rapid progression: eGFR decline > 5 mL/min/1.73m<sup>2</sup> per year OR > 25% decline within 12 months
- Proteinuria > 500–1000 mg/g ACR unexplained or non-responding to treatment
- Hypertension refractory to 3 agents at maximum dose (include diuretic)
- Persistent hyperkalaemia, anaemia unresponsive to iron, or CKD-MBD complications
- Suspected hereditary kidney disease (ADPKD, Alport syndrome, FSGS) or systemic disease
- Glomerular haematuria or red cell casts on urine microscopy

DISCLAIMER: This document is intended for use by licensed medical professionals in clinical and educational settings. Drug doses and protocols are indicative; always verify against current BNF, local formulary, and individual patient factors. Clinical guidelines evolve — confirm against latest KDIGO/NICE/AHA guidance. This document does not replace specialist nephrology consultation for complex or high-risk patients.